Abdelhak Kadouma

Software Engineer | AI/ML Engineer | Cloud-Edge Systems
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Professional Summary

Software Engineer and doctoral researcher with over three years of experience developing backend services, REST APIs, distributed applications, data-processing pipelines, and AI/ML systems. Strong background in Python and Java with hands-on experience in FastAPI, Spring Boot, Django, SQL and NoSQL databases, Docker, Kubernetes, and production monitoring. Experienced in turning research ideas and early prototypes into maintainable software through backend integration, reproducible experimentation, containerised deployment, observability, and system integration across cloud-edge environments.

Technical Skills

Programming: Python, Java, SQL, JavaScript, TypeScript
Backend and APIs: Spring Boot, FastAPI, Django, Django REST Framework, REST APIs, microservices, asynchronous processing, system integration
AI and Machine Learning: scikit-learn, PyTorch, supervised learning, time-series forecasting, drift detection, feature engineering, model evaluation, LLM integration
Data and Databases: Pandas, NumPy, PostgreSQL, MySQL, MongoDB, Redis, SQLite, data ingestion, transformation, validation
Cloud and Infrastructure: Docker, Kubernetes, Linux, Nginx, Git, CI/CD, containerised deployment, cloud-native systems
Reliability and Observability: Prometheus, Grafana, automated testing, health checks, structured logging, runtime metrics, debugging and failure analysis

Professional Experience

University of Vaasa Jun. 2026 – Present
Doctoral Researcher *Vaasa, Finland*

- Develop Python-based machine-learning and data-processing pipelines for workload forecasting, predictive application management, and analysis of operational cloud telemetry.
- Build reproducible workflows covering data preparation, feature engineering, model training, chronological validation, statistical analysis, benchmarking, and error analysis.
- Translate research methods into reusable software components for evaluating adaptive resource-management approaches in distributed cloud-edge systems.

Fingletek Oy Sep. 2023 – Present
Software Engineer and Research Collaborator *Espoo, Finland*

- Design and develop Python and Java backend services, REST APIs, microservices, data-processing workflows, and integration components for distributed cloud-edge platforms.
- Build software for application lifecycle management, deployment automation, monitoring, semantic metadata processing, and AI/ML-enabled workflows.
- Deploy and operate containerised services using Docker and Kubernetes with relational and NoSQL databases, Nginx, Prometheus, and Grafana.
- Improve reliability through automated testing, health checks, structured logging, runtime metrics, and investigation of API, database, networking, configuration, and service-integration failures.
- Collaborate with engineers, researchers, and project stakeholders to define requirements, evaluate technical alternatives, implement features, integrate systems, and document reusable software components.

Fingletek Oy Sep. 2022 – Sep. 2023
Software Engineer Intern *Espoo, Finland*

- Developed backend services, REST APIs, and distributed application-management functionality using Python, Django, React, Docker, and Kubernetes.
- Contributed to application deployment, integration, debugging, and monitoring workflows across cloud and edge environments.

Selected Projects

DynAPM – Predictive Application Profile Maintenance 2025 – 2026

- Developed a reproducible ML pipeline combining workload forecasting and online drift detection using traces from 1,576 production virtual machines; improved CPU and memory profile-class accuracy by 13.0 and 14.5 percentage points respectively and reduced SLA-risk events by 28%.

TimeTrack – Multi-Metric Forecasting Framework 2025 – 2026

- Developed a reproducible framework for forecasting CPU, memory, disk, and network telemetry, including data preparation, model training, chronological validation, benchmarking, and error analysis.

LLM-Based Application Profile Generator 2025 – 2026

- Built a FastAPI service that transforms natural-language application requirements into validated structured profiles using an LLM, with asynchronous processing, persistent job tracking, output validation, and integration with deployment workflows.

Avaruus – Cloud-Edge Application Management Platform 2024 – 2026

- Developed backend APIs and orchestration components for deploying, monitoring, and managing distributed applications across Kubernetes-based cloud-edge environments with integrated databases and observability.

OSR – Semantic Deployment Integration 2023 – 2026

- Developed integration software connecting semantic application metadata, catalogues, and cloud-edge lifecycle workflows, contributing to the successful completion of the EU-funded AC3 project.

Education

University of Vaasa 2024 – Present
Doctoral Studies in Computer Science *Vaasa, Finland*

Higher School of Computer Science 2017 – 2023
State Engineering Degree and Master's Degree in Computer Science *Algiers, Algeria*

Languages

Arabic: Native | **English:** Professional working proficiency | **French:** Professional working proficiency